SSLC SURE A Plus Notes

Social Science

Towards Sustainability

1.Do you think that resources are simply free gifts of nature?

⁂ That's not entirely true. Resources are also formed as a result of human efforts.

⁂ Therefore, humans are an essential element in the formation of resources.

⁂ We convert the materials available in our environment into resources and use them to meet our needs.

2.Do you think human a resource?

⌘ Humans can be considered as resources because they create and develop resources using their abilities, skills, and technology.

3.Resources can be classified into two categories: Natural resources and Man-Made resources

Natural Resources	Man-made Natural Resources
Air, water, soil, sunlight, minerals, fossil fuels, plants, and animals	Buildings, roads, machines, electronics, plastic, paper, furniture, vehicles

4.Resources can also be classified into different categories based on their origin, renewability, ownership, and state of development.Based on the origin, resources are classified into two types:Make a table

Biotic Resources	Abiotic Resources
Anything that is part of the biosphere and can be used by humans is called a biotic resource. Humans, plants, and animals are examples of biotic resources.	Resources that are made of non-living matter are called abiotic resources. Air,Soil, Water, Minerals are examples.

5.Based on their replenishment capacity or renewability, resources can be classified into two types: Renewable resources and Non-renewable resources.

Renewable Resources	Non renewable resources
Continuously produced or replenished in nature.	Non-renewable resources are those that have been formed

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Many of them do not diminish after use and can be reused. These resources are considered sustainable as long as they are not overused. Most renewable resources are readily available to humans. Eg.Solar, Wind, Biomass, Water	over millions of years and whose reserves decrease with use. Minerals and fossil fuels are examples of non-renewable resources. Some of these, like metals, can be reused, but others, like fossil fuels, deplete with use and cannot be reused.

6.Resources can also be classified into different types based on their ownership. Make a flowchart (SCERT Model)

Resources			
Personal Resources	**Community Resources**	**National Resources**	**International Resources**
Owned by an individual Eg: House, Yard, Own vehicle etc	Accessible to all members of the society Eg: Public play ground, Parks, cemeteries etc.	Owned by the nation.Mineral deposit, forest, wildlife, water resources, land with in the countries border, the ocean from the coast up to 12 nautical miles.	Owned by international agencies. Eg: Marine resources beyond 200 nautical mile

7.Are all the resources on Earth accessible to us?

⌘ Due to limitations in technology or the lack of appropriate technological advancement, we are not able to develop many things on Earth into resources.

⌘ Sometimes, we have to wait until the adequate technology is developed to convert them into resources.

8.Based on the state of development, resources can be classified into two types: Potential resources and Developed resources.Examine (SSLC Pre Model)

Potential Resources	Developed Resources
Potential Resources are resources that are found in a region but are not yet fully utilized. For example, Rajasthan and Gujarat have	Developed Resources are resources whose quantity and quality have been surveyed, assessed, and adequate technology

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a huge potential for generating energy from wind and solar power, but these sources are currently only marginally used for power generation.	has been developed to utilize them efficiently. Eg.Coal, Petroleum, Natural gas

9.Is petroleum a developed resource?

⌘ While petroleum as a whole is a developed resource, specific deposits might still be "potential" if they are in remote areas where extraction technology is not yet cost-effective.

10.What do you mean by stocks? **(SSLC Pre Model)**

℮ Materials that can meet human needs but for which humans do not have the appropriate technology to utilize are called stocks.

℮ Hydrogen can be utilized as a rich source of energy, but we have not yet developed the advanced technology to use it as a resource today.

℮ There are many energy sources in nature that we may be able to utilize in the future.

11.Define Natural resources?

◊ Natural resources are substances, objects, or forms of energy that are obtained from nature and are useful to humans.

◊ They are valuable resources available in their natural state and are not created by humans.

12.Natural resources such as mineral resources and energy resources are indispensable to human life. Elucidate

^ Natural resources include air, water, soil, minerals, plants, animals, fossil fuels, and more.

^ We depend on these resources to meet our domestic, agricultural, industrial, and energy needs.

^It influences our daily life much.

13.What you meant by Minerals? Explain it's importance

> Minerals are naturally occurring organic or inorganic substances with definite chemical and physical properties.

>They are chemical compounds in which atoms are arranged in a specific manner.

> Most minerals are formed by the combination of two or more elements, while only a few elements occur in their pure (elemental) state in nature.

>The welfare of the people, the development of science and technology, and the economic growth of a country depend, to some extent, on the availability of minerals.

14.Minerals can be classsified into Ferrous minerals and Non Ferrous minerals

Ferrous minerals- iron ore (hematite, magnetite), manganese, and chromite

Non Ferrous minerals-gold, copper

Non-Metallic Minerals can be classsified into Inorganic minerals and organic minerals

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Inorganic minerals-quartz, feldspar, mica, calcite, elements like iron, gold, copper
organic minerals-mellite, humboldtine, fichtelite, and idrialite

15.Write a short essay on The Versatility and Industrial Utility of Minerals

⚹ Metals and non-metals, along with their various compounds, are indispensable to modern industry, technology, and medicine.

⚹ While iron remains a staple for general manufacturing, titanium is prized for its high strength-to-weight ratio in aerospace engineering.

⚹ Gold serves critical roles beyond aesthetics, particularly in high-tech electronics.

⚹ Furthermore, minerals like manganese support the pharmaceutical sector, while non-metallic minerals provide the essential raw materials necessary for diverse industrial applications.

16. Elucidate the economic extraction of metals from Ores

☞ The Earth's crust contains various rocks rich in metallic minerals that can be harvested economically.

☞ These materials, known as ores, are defined by the ability to extract metals from them for a profit.

☞ A single metal, such as iron, may be sourced from multiple minerals like hematite or magnetite; conversely, a single ore deposit can often yield several different types of metals.

17.What is mineral deposits ?

⌑ When beneficial minerals found as natural occurrences in unusually high concentrations, they can be termed as mineral deposits.

18.Are all mineral-rich areas considered deposits?

₹ No, not every area containing minerals is considered a "deposit.

₹ " In geology and mining, there is a distinct hierarchy based on concentration and economic potential.

19.Write a seminar paper on mineral deposits that have greatly contributed to the progress of human society across the world.

Iron: The Foundation of Modern Civilization

$ Iron is a globally indispensable metal that has driven human progress since the Iron Age.

$ Today, the largest reserves are found in Australia, Brazil (home to the massive Carajás Mine), and Russia, with India also ranking as a top global producer.

$ Its versatility makes it essential for everything from small household items to massive industrial infrastructure.

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$ Because of its strength and durability, iron is used in:

- Infrastructure,Transportation,Household Items

$ Iron is a key component in the manufacturing of steel. About

- 98% of the iron ore mined worldwide is used to produce steel.
- Nearly 50 countries are involved in iron ore mining. Steel
- plays a crucial role in maintaining a strong industrial base

Copper: From Ancient Tools to Modern Technology

$ Copper is one of the oldest metals used by humanity, marking the transition from the Stone Age to the Copper Age.

$ Because it often occurs in a pure metallic state in nature, it was easily shaped by early humans into tools, weapons, and ornaments.

$ Today, its exceptional electrical and thermal conductivity has made it a cornerstone of modern infrastructure, with Chile leading the world in its production through massive operations like the Escondida and Collahuasi mines.

Industrial Metals and Their Diverse Applications

$ Beyond iron and copper, several other metals are essential to modern manufacturing and infrastructure.

$ Aluminium is valued for its versatility in transportation, construction, and electronics, while Manganese and Chromium are critical agents in refining and strengthening steel.

$ Additionally, Lead remains a fundamental component in energy storage and industrial ceramics.

20.What are energy resources? Define fuel. Write an example of renewable fuel. (SCERT model)

☞ Any substance that can produce heat, move objects, or generate electricity is called an energy source.

☞ A substance that releases energy through a chemical process is known as a fuel.

21.List the energy sources you are familiar with:

¢ Solar Energy,Coal,Hydel (Hydroelectric) Power,Petroleum,Natural Gas

22.When do energy sources become resources?

✪ In simple terms, an energy source is the raw natural phenomenon (like sunlight or flowing water), while an energy resource is the usable supply of that energy that

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humans can actually harness.

✪ Energy resources play a crucial role in human life and in the economy of every country.

23.Energy resources can be classified into two categories based on their renewability

Non-Renewable resources/Conventional resources	Renewable resources/Non- conventional resources
Natural gas,coal, petroleum	Solar energy, hydroelectricity,tidal energy, geothermal energy, wind, nuclear energy, bioenergy

24.The characteristics of both conventional and non-conventional energy resources (SSLC Pre Model)

Non-Renewable resources/Conventional resources	Renewable resources/Non- conventional resources
* These resources get depleted and exhausted with use. * They cause pollution. * Very expensive to maintain, extract, and transport	* Available in plenty * Eco-friendly * Usually pollution-free

25.Why humans rely on resources ?

⊙ The earth has provided humankind with plenty of energy resources. ⊙ Humans rely on these resources for food production, transportation, healthcare, and almost every sector of human activity.

25. What you meant by Fossil fuels ? List the major fossil fuels

₩ Fossil fuels belong to the category of non-renewable resources, but they are very important sources of energy.

₩ Fossil fuels are formed from the fossilized remains of ancient plants and animals.

₩ Eg-Natural gas,coal, petroleum

26.Briefly explain Coal

⁂ Coal is a combustible sedimentary rock formed from ancient plant remains and is distinguished by its high carbon content and dark coloration.

⁂ While widely recognized as a primary fuel for thermal power generation, coal is also critical in metallurgy for extracting and refining metals.

⁂ Beyond energy, it serves as a versatile chemical raw material in the production of diverse goods, including medicines, pesticides, and dyes.

27.Prepare an article on Petroleum:

⌘ The Evolution and Impact of "Black Gold" Petroleum, or "rock oil," is a

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complex liquid mixture that has evolved from a sacred ancient phenomenon into the world's most influential energy source.

⌘ Historically used by Egyptians for mummification and as a waterproofing agent, it became the primary driver of the Industrial Revolution, eventually surpassing coal.

⌘ Today, its economic power is best demonstrated by nations like Saudi Arabia, which transformed into a developed global power through its vast oil reserves.

⌘ Naturally occurring petroleum is extracted as crude oil, which varies in color from black to yellow.

⌘ It must be processed in a refinery to separate its components into usable products through a process called fractional distillation.It uses as Energy and Fuels,Petrochemicals

Global Economic Influence

⌘ The availability of low-cost petroleum from regions like West Asia has fundamentally altered global trade and lifestyle.

⌘ Because it powers almost every sector of modern society—from transportation to manufacturing—it is aptly nicknamed "Black Gold."

28.Prepre a discussion note on the topic, 'Uncontrolled Consumption of Fossil Fuels and the Energy Crisis',

The Impending Fossil Fuel Crisis

e The rising global demand for energy is rapidly depleting non-renewable fossil fuel reserves, which include coal, petroleum, and natural gas.

e Since coal's early use in metallurgy and steam engines, followed by the 19th-century commercialization of petroleum and the current domestic reliance on natural gas, these finite resources have powered modern development.

e However, their uncontrolled consumption threatens a total exhaustion of supplies.

e Without these essential energy sources, global developmental activities face a potential standstill and a catastrophic energy crisis.

29.What are the importance of wise energy use ?

To address the imminent energy crisis and the environmental damage caused by over-exploiting fossil fuels, we must pivot toward non-conventional energy sources.

These are sustainable, naturally replenished, and abundant alternatives that offer a path for long-term development without harming nature

30.Examine the major non-conventional energy sources and see how they can be utilised

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to meet our energy needs. **(SSLC Pre Model)**

Or Hold a discussion in what ways can non-conventional energy resources be utilised in human life? **(SSLC Pre Model)**

Or What energy can be used where petroleum and coal are insufficient **(SSLC Model)**

Or A thermal power plant causes atmospheric pollution due to the intensive use of coal for generation of electricity. Suggest eco friendly alternative methods of generating electricity and explain the characteristics of each **(SSLC 2026)**

Solar Energy	? The Sun serves as our planet's primary energy source. ? To utilize this power, modern technology employs two main methods: photovoltaic and solar thermal capture. ? While thermal methods focus on heat, photovoltaic cells convert solar radiation directly into electrical energy, which can be stored and used for a wide range of applications.
Wind energy	?The kinetic energy of the wind is converted into electricity. ?The force of the wind is used to spin a windmill, which in turn drives a generator connected to it, producing electricity.
Geo thermal energy	? The heat released when magma erupts from the Earth's interior to the surface is efficiently stored and converted into electrical energy. ? In addition, it is possible to generate thermal energy from the hot water and vapour that emerge from the Earth through geysers. ? It has significant importance as an alternative energy source.
Tidal Energy	? The movement of ocean tides is used to spin turbines and generate electricity.
Bio energy	? Bioenergy is energy produced from organic matter and organic waste. ? It helps improve the quality of life for rural populations in developing countries and reduces environmental pollution. ? Today, projects to produce energy from urban waste are also being implemented.

31.Energy Conservation Day is observed in December 14

32.Why do we observe Energy Conservation Day

Or What are the importance of Energy Conservation Day

§ As natural resources dwindle, balancing consumption with development is vital to preserve the Earth's self-regenerative capacity.

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§ True conservation involves protecting ecosystems and the global climate by transitioning from finite, harmful fossil fuels to renewable energy sources like solar, wind, and geothermal power.

§ Sustainable development—the careful management of resources to prevent depletion—is essential to ensure that future generations inherit a viable planet.

33.Write a short note on the topic: "The Role of Children in Sustainable Development."
Or Examine the role of the conservation of resources in sustainable development. (SCERT Model)

◊ Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

◊ We can move forward by engaging in activities that aim to achieve the objectives of sustainable development, protect the Earth, and conserve resources for a better tomorrow.

◊ We must try to develop sense of usage of energy sources and other resources such as solar energy, wind energy, hydroelectricity, and geothermal energy, simultaneously reducing the use of fossil fuels, which are scarce, costly to replenish, and have adverse environmental impacts.We can lead campaign for

◊ 3Rs,Energy saving and conservation.Assemblies,club activities, public awareness programmes,surveys can be arranged.

34.“The world has enough for everyone's need, but, not enough for everyone's greed" Who said it?

^ Mahatma Gandhi

35.Elucidate the importance of Hydrogen a non-conventional energy resource. (SCERT Model) (SSLC Pre Model)

> Hydrogen is utilized as a non-conventional energy source.

> Hydrogen, is used to produce energy, which can serve as a clean fuel.

> When hydrogen is burned in a fuel cell, only water vapor is reused.

> It does not release pollutants like fossil fuels when they are burned.

>This energy is useful in areas ranging from transportation to various economic activities. It can be used.

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All the Very best dear students

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